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RESEARCH ARTICLE

Routing Optimization for Large-Scale Inspection-Maintenance of EV Charging Spots With Multiple Types of Personnel and Composite Customer Satisfaction: A Multi-Stage Approach

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ABSTRACT Electric vehicle (EV) charging spots require frequent inspections to identify potential safety hazards, and charging piles in charging spots exhibit a significant failure rate, necessitating maintenance. The efficiency and quality of inspection-maintenance are crucial components in the operation and management of charging spots, directly affecting the operation cost, service level, and user experience. The inspection-maintenance of EV charging spots requires two different types of personnel, and the impact on customer satisfaction cannot be ignored. In addition, the large-scale charging spots have further increased the difficulty of routing. A novel routing optimization problem for large-scale inspection-maintenance of EV charging spots is proposed in this paper, considering multiple types of personnel and composite customer satisfaction. While inspections can be carried out by general personnel, maintenance requires professional engineers. Composite customer satisfaction is with respect to the waiting time for maintenance of faulty charging spots and the duration of absence of professional engineers in the working station. A bi-objective routing optimization model is proposed, minimizing the cost of arranging personnel and traveling between charging spots, and the loss of composite customer satisfaction simultaneously. Due to the large scale of the problem caused by the enormous number of charging spots, an effective multi-stage solution approach is proposed. Finally, numerical experiments, managerial insights, and a case study are provided. The purpose of this research is to help the operators to decide the routes for inspection-maintenance of EV charging spots and offer valuable guidance in balancing operational efficiency with customer satisfaction.

INDEX TERMS Routing optimization, large-scale inspection-maintenance, multiple types of personnel, composite customer satisfaction, multi-stage approach.

I. INTRODUCTION

With the continuous growth of environmental pressure and the depletion of fossil fuels, reducing carbon emissions has

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emerged as a key measure for the global response to climate change [1]. Many countries and regions have made proactive efforts to reduce carbon emissions. For example, the Chinese government has pledged to reach peak carbon emissions by 2030 and gradually reduce emissions to achieve carbon neutrality by 2060 [2]. At the recent COP 28 UN climate change

conference, more than 150 heads of state and government committed to accelerating action across all areas by 2030 [3]. The transportation sector constitutes one of the main sources of carbon emissions, and switching to electric vehicles (EVs) is an important step towards a low-carbon and environmentally friendly future [4]. Unlike vehicles that rely on fossil fuels, EVs utilize electricity as their energy source. Although this shift can effectively improve environmental friendliness, it also brings the issue of “range anxiety” for drivers [5]. To alleviate this problem, many countries and regions have constructed extensive charging systems with numerous EV charging spots across wide areas [6].

EV charging spots, akin to gas stations, are equipped with multiple charging piles. However, charging piles differ from gas pumps in gas stations in several aspects. Firstly, as electrical equipment, charging piles require frequent inspections to identify potential safety hazards [7]. While the inspection process is relatively simple-involving observing panel displays, listening for abnormal sounds, and checking for high temperatures-the sheer number of charging piles poses significant challenges for inspection. For example, by the end of 2023, China alone had 8.596 million charging piles. Secondly, charging piles exhibit a high failure rate [8]. In China, more than ten percent of the constructed charging spots contain faulty charging piles, creating additional challenge for EV charging spot operators. Consequently, operators need not only build charging spots but also arrange for professional engineers to carry out maintenance tasks [9]. In summary, these practical situations require that EV charging spot operators establish working stations in their responsible areas and arrange personnel to conduct inspections and maintenance.

The efficiency and quality of inspection-maintenance are crucial components in the operation and management of charging spots, directly affecting the operation cost, service level, and user experience. From a cost perspective, operating charging spots is characterized by high initial investments and long return periods. According to statistics from Star Charge, a large EV charging spot operator in China, the operation and maintenance costs account for about 30%-50% of the total cost of running charging spots. Meanwhile, the initial investment in building charging spots may take more than 3 years to recover. Optimizing the routing for inspection-maintenance can significantly reduce travelling costs, labor costs, and other associated expenses during the inspection process, thereby reducing the overall operating costs of charging spots. From a service level perspective, efficient inspection-maintenance process can reduce downtime, lower the likelihood of customers encountering faulty charging spots, ultimately improve user experience and improve competitiveness in the market. Given these critical cost and service considerations, the routing optimization problem for the inspection-maintenance of EV charging spots emerges as a key challenge for operators.

Planning routes for the inspection-maintenance of EV charging spots presents several unique characteristics. First,

EV charging spot operators need to conduct both inspection and maintenance tasks simultaneously, but with different personnel requirements. While inspection can be carried out by general personnel, maintenance demands the expertise of professional engineers [10]. In addition, the time required for inspection and maintenance differs, so do the costs associated with the general personnel and professional engineers. Consequently, the routing optimization problem becomes complex, requiring consideration of diverse skill sets, varying task durations, and differing cost structures. Second, due to the widespread deployment of EV charging spots, the number of charging spots in each region can be substantial [11]. Take Xi'an, a Chinese city, as an example. There are as many as 1,281 charging spots. The large-scale distribution compounds the complexity of the routing optimization, making it challenging to determine the most efficient inspection-maintenance plan.

In addition to the challenges of managing multiple types of personnel and large-scale operations, customer satisfaction emerges as an important factor in EV charging spot management. Our interviews with several large Chinese EV charging spot operators, such as TELD and Star Charge, reveal that they prioritize customer satisfaction alongside operational costs when planning inspection-maintenance arrangements. Two main sources can lead to losses in customer satisfaction. Firstly, meeting customer needs is a key factor for competitive advantage, and non-functional charging piles due to lack of maintenance can lead to significant customer satisfaction losses [9], [12]. Therefore, operators need to prioritize swift repairs of faulty charging spots in their inspection-maintenance plan. Second, the working station established by operators also serve as customer service points. Customers often contact these working stations with queries or issues related to charging piles, such as usage instructions, unresponsive charging plugs, or slow charging speeds. Ideally, professional engineers can address these concerns promptly and effectively. However, when engineers are absent due to maintenance duties, regular telephone operators may struggle to provide satisfactory solutions. Research has shown that customer service performance affects the customer satisfaction [13], [14]. The absence of professional expertise can lead to delay in problem resolution and subsequent customer dissatisfaction. The operators need to consider how their inspection-maintenance routing strategies affect both aspects of customer satisfaction while managing operational costs effectively.

We propose a novel routing optimization problem for large-scale inspection-maintenance of EV charging spots that incorporates multiple types of personnel and composite customer satisfaction. This approach addresses the unique challenges of simultaneous inspection and maintenance requirements, the involvement of diverse personnel, and the importance of customers in the EV charging industry. Although there is already a significant body of work on routing optimization for inspection, these studies mainly

focus on power lines [15], [16], [17], [18]. The core objective of these studies is to traverse all power lines with the lowest possible cost or in the shortest possible time. However, research that considers both inspection and maintenance simultaneously is limited, and the impact of inspection-maintenance on customer satisfaction is often overlooked. Another problem that bears the closest resemblance to our proposed model is the Multiple Traveling Salesman Problem (MTSP), which plans routes to traverse all vertices in a network. However, MTSP research does not consider that inspection-maintenance requires multiple different types of personnel, and cannot directly deal with the routing optimization problem for inspection-maintenance with multiple types of personnel.

This paper investigates the routing optimization problem for large-scale inspection-maintenance of EV charging spots with multiple types of personnel and composite customer satisfaction, aiming to provide charging spot operators with a routing plan for inspection-maintenance that effectively balances cost and customer satisfaction. Our contributions can be summarized as follows. (1) We propose a novel routing optimization problem for the inspection-maintenance of EV charging spots. Unlike previous related studies, we have considered multiple types of personnel and composite customer satisfaction. To address these complex issues, we develop a bi-objective model that simultaneously minimizes the total cost and the loss of composite customer satisfaction. (2) Based on the characteristics of the large-scale inspection-maintenance of EV charging spots, we propose a multi-stage approach. This includes a clustering method to reduce the scale of the proposed problem, a fast greedy approach to estimate the cost and time for each cluster, and an adaptive Nondominated Sorting Genetic Algorithm-II (NSGA-II) to obtain the Pareto frontier. (3) Through sensitivity analysis, we have uncovered several key insights. Experienced professional engineers can improve inspection-maintenance efficiency, even with higher wages. The number of personnel is crucial, and it is crucial to ensure a sufficient workforce. It is beneficial to reduce travel time between charging spots, unless it leads to increased traveling costs. The weight assigned to customer satisfaction has a notable impact, but its optimal value needs to be determined based on specific circumstances. This research can provide the model and corresponding solution approach to help the operators to decide the routes for inspection-maintenance of EV charging spots with multiple types of personnel, and offer valuable guidance in balancing operational efficiency with customer satisfaction.

The remainder of this paper is organized as follows. Section II provides a thorough literature review. Section III introduces the problem definition and develops the mathematical model. Section IV details a multi-stage approach. Section V presents numerical results. Finally, Section VI concludes the paper. The block diagram summarizing the contents and the flow of subsequent sections is shown in Figure 1.

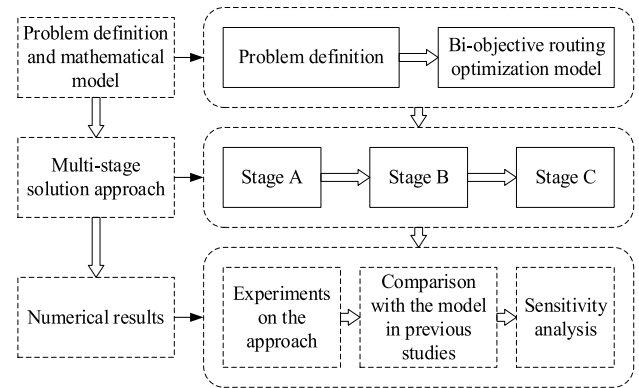


FIGURE 1. Block diagram summarizing the contents and the flow of the subsequent sections.

II. LITERATURE REVIEW

Although substantial research has conducted on the routing optimization for the inspections, the majority of these studies have focused on power lines. These studies mainly focus on planning routes that traverse all power lines. For instance, Liu et al. proposed a grid-based inspection strategy, developing a mathematical model solved using a genetic algorithm [15]. Liu et al. studied a two-layer routing problem for power line inspection using cooperating vehicles and drones, with the objective of minimizing total completion time [16]. Nekovář et al. investigated a multi-tour routing problem in power line inspection planning. They employed a single objective of minimizing costs related to total length of all routes and adapted an adaptive tabu search algorithm to solve their model [17]. Li et al. addressed a routing problem under multi-wind field conditions and introduced a novel honey badger-fruit fly algorithm to solve it [18].

Given the relevance of the MTSP to the problem addressed in this paper, a review of relevant research is provided. Research on MTSP primarily aims to minimize the total length of all routes. Various approaches have been proposed to tackle this problem efficiently. Zhang et al. introduced a state transition simulated annealing algorithm for MTSP with closed routes, demonstrating stronger local search capability and better approximation of optimal solutions compared to other methods [19]. Ergüven and Polat developed a relative distances approach that enabled real-time scheduling for MTSP [20]. Wang et al. investigated MTSP with time windows and multiple depots, proposing an improved algorithm combining partheno-genetic algorithm with the invasive weed algorithm. Their approach effectively avoided local convergence, showed reduced parameter sensitivity, and exhibited good stability [21]. Jiang et al. proposed a hybrid colony-partheno genetic algorithm for large-scale MTSP with multiple depots, which performed well in solving large-scale MTSP [22]. He and Hao developed a unified population-based memetic search algorithm for both single and multiple depots MTSP [23]. Tong et al. developed a highly robust hybrid artificial bee colony algorithm for the

MTSP with multiple depots [24]. Bérczi et al. considered a variant of MTSP, the many-visits MTSP, and proposed a polynomial-time approximation algorithm [25]. Cunha et al. presented a novel variant of the MTSP family and developed an adaptive large neighborhood search algorithm to address it [26].

While the aforementioned studies primarily focus on the MTSP, recent research has increasingly explored its application to real-world scenarios. Cornejo-Acosta et al. investigated a depot-free MTSP based on real operational scenarios, aiming to minimize operational costs, represented by the total length of all routes. To solve this problem, they developed a quadratic programming model and utilized the optimization software Gurobi [27]. Hu et al. proposed a reinforcement learning approach to solve a realistic MTSP over graphs. Their experimental results indicated that reinforcement learning significantly outperforms traditional methods when dealing with complex, large-scale problems [28]. Garn explored the challenge of dynamic customers in practical scenarios, developed a continuous approximation model for MTSP, and proposed a balanced dynamic heuristic for its solution [29]. Liang et al. developed a bipartite graph neural network with an attention mechanism for solving MTSP, demonstrating the potential of advanced machine learning techniques in tackling urban logistics challenges [30].

While many MTSP studies focus on the total length of all routes, a significant body of research has explored an alternative objective: minimizing the length of the longest route. Mahmoudinazlou and Kwon made a notable contribution on this area by proposing a hybrid genetic algorithm with the single objective of minimizing the longest route. Their results demonstrated the algorithm's effectiveness in both optimal and average performance metrics [31]. Liu et al. applied the longest route minimization objective to MTSP involving unmanned vehicles. They developed a mathematical model and introduced a genetic simulated annealing algorithm to solve it [32]. Kloster et al. further incorporated drones into the MTSP framework, developed a mixed-integer linear programming model and proposed decomposition-based metaheuristic algorithms [33].

While many studies focus on single-objective optimization in MTSP, there is growing interest in multi-objective approaches that offer more nuanced solutions to complex routing problems. In the bi-objective MTSP, the minimization of the total length of all routes is typically treated as the upper-level objective, and the minimization of the longest route is considered as the lower-level objective. Karabulut et al. developed a linear programming model for the bi-level MTSP and proposed an evolutionary strategy to solve it [34]. Zheng et al. introduced a two-stage heuristic to solve the bi-level MTSP. Their analysis demonstrated that the algorithm possesses high local search efficiency and effectively avoids local optima [35]. He and Hao proposed a hybrid algorithm with neighborhood reduction for solving the bi-level MTSP, and their numerical results showed strong competitiveness [36].

The key points of the related work mentioned above are summarized in Table 1. From Table 1, it is evident that existing research related to power line inspection primarily focuses on planning routes that traverse all power lines. The core objective of these studies is to traverse all power lines with the lowest possible cost or in the shortest possible time. Existing research related to MTSP primarily focuses on minimizing the total length of all routes as a single objective. There is a growing body of research on bi-level MTSP, which considers both the minimization of total longest route and the minimization of the longest route. Recent studies have begun to explore more complex scenarios, including the incorporation of unmanned vehicles and drones, as well as the consideration of dynamic elements such as customer locations. While these works provide a solid foundation for the in-depth exploration of MTSP, they do not fully address the specific challenges presented in this paper.

To the best of our knowledge, no existing research in the literature effectively addresses the specific issues raised in this paper. Our study distinguishes itself from previous work in several aspects. First, in terms of costs, while related studies primarily consider minimizing the total length of all routes, the problem in this paper also includes the fixed cost associated with personnel arrangements. Second, we introduce the concept of composite customer satisfaction, which is not typically addressed in tradition MTSP research. Third, our study uniquely incorporates the practical challenges of operating EV charging spots, which involve both inspection and maintenance requirement. This scenario necessitates the consideration of two different types of personnel, a complexity not addressed on related studies. These distinctive features set our research apart from existing work in the field. We will develop a bi-objective model based on MTSP that comprehensively considers the characteristics of the inspection and maintenance of EV charging spots. Subsequently, we will also propose a multi-stage approach to effectively manage the complexity of the problem.

III. PROBLEM DEFINITION AND MATHEMATICAL MODEL

A. PROBLEM DEFINITION

Based on the actual road network topology, an undirected complete abstract network can be obtained. The vertices on this network include three types: a working station established by the EV charging spot operator, a series of charging spots requiring inspection, and a series of charging spots requiring maintenance. The operator can arrange two types of personnel from the working station to complete inspection-maintenance tasks: general personnel capable of conducting inspection work, and professional engineers skilled in performing both inspection and maintenance tasks. Both types of personnel incur travel costs between vertices. However, the fixed costs for arranging these personnel differ, with professional engineers commanding higher rates than general personnel. Figure 2 is a schematic diagram of the inspection-maintenance process. The inspection-maintenance process

TABLE 1. Summary of the related work.

Source	Objectives					Requirements for routes	Additional consideration	Type of service object	Type of vehicle / personnel
	MTL	MLR	BLO	MTC	MCS				
MTSP									
Zhang et al. (2021)	√					traverse vertexes		1	1
Ergüven et al. (2024)	√					traverse vertexes		1	1
Wang et al. (2020)	√					traverse vertexes	with time windows	1	1
Jiang et al. (2020)	√					traverse vertexes	with multiple depots	1	1
He et al. (2023)	√					traverse vertexes	with multiple depots	1	1
Tong et al. (2024)	√					traverse vertexes	with multiple depots	1	1
Bérczi et al. (2023)	√					traverse vertexes	many-visits	1	1
Cunha et al. (2024)	√					traverse vertexes	subsets called families	1	1
MTSP with specific application scenario									
Comejo-Acosta et al. (2023)	√					traverse vertexes	depot-free	1	1
Hu et al. (2020)	√					traverse vertexes	over graphs	1	1
Garn (2021)	√					traverse vertexes	with dynamic customers	1	1
Liang et al. (2024)	√	√				traverse vertexes	urban logistics	1	1
MTSP with the objective of MLR									
Mahmoudinazlou et al. (2024)		√				traverse vertexes		1	1
Liu et al. (2023)		√				traverse vertexes	with unmanned vehicles	1	1
Kloster et al. (2023)		√				traverse vertexes	with drone stations	1	2
Bi-level MTSP									
Karabulut et al. (2021)			√			traverse vertexes		1	1
Zheng et al. (2022)			√			traverse vertexes		1	1
He et al. (2022)			√			traverse vertexes		1	1
Routing optimization for inspection of power lines									
Liu et al. (2018)	√					traverse arcs	based on grid cooperated vehicle and drone	1	1
Liu et al. (2019)	√					traverse arcs		1	2
Nekovář et al. (2021)	√					traverse arcs	multi-tour	1	1
Li et al. (2024)	√					traverse arcs	under multi- wind field	1	1
This paper				√	√	traverse vertexes	with service time / limited number of personnel	2	2

MTL = minimizing the total length of all routes
MLR = minimizing the length of the longest route
BLO = bi-level objectives
MTC = minimizing the total cost comprising multiple factors
MCS = minimizing the loss of composite customer satisfaction

involves several time-related factors. All personnel spend time traveling between vertices, as well as inspecting or maintaining charging spots. Notably, maintenance tasks require more time than inspections. Waiting for maintenance of faulty charging spots can lead to the loss of customer satisfaction related to time. If all professional engineers are absent from the working station, it may result in the loss of customer satisfaction. The objectives of the problem are to minimize the total cost, which comprises the fixed costs of arranging personnel and the travel costs between charging spots, and to minimize the loss of composite customer satisfaction.

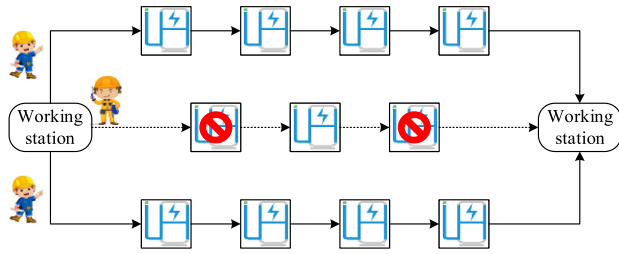


FIGURE 2. Schematic diagram of the inspection-maintenance process.

1) ASSUMPTIONS

(1) Each charging spot is visited only by one type of personnel—either general personnel or professional engineers.

(2) The number of general personnel and professional engineers that can be arranged to the working station is subject to specific constraints.

(3) The loss of customer satisfaction due to maintenance waiting time is contingent upon the total waiting time for maintenance of all faulty charging spots.

(4) The duration for which the working station lacks a professional engineer is contingent upon the earliest return time of any professional engineer to the working station.

2) PARAMETERS AND DECISION VARIABLES

The parameters and decision variables involved in the proposed problem are systematically summarized. Table 2 presents the descriptions of all the parameters. The decision variables encompass the following types:

x_{ij}^p : 0-1 decision variable, denoting whether personnel p passes arc aij ;

y_i^p : 0-1 decision variable, denoting whether personnel p inspects or maintains charging spot i ;

v^p : 0-1 decision variable, denoting whether the route of personnel p includes at least a charging spot that requires maintenance;

t_i^p : non-negative decision variable, denoting the departure time of personnel p from vertex i .

B. BI-OBJECTIVE ROUTING OPTIMIZATION MODEL

The aforementioned problem encompasses two objectives. The first objective is to minimize the total cost, while the second objective is to minimize the loss of composite customer

satisfaction. Regarding the cost objective, we can use C_1 to represent the fixed cost of arranging personnel (both general and professional engineers) from the working station, and C_2 to denote the traveling cost associated with the routes taken by these personnel for inspection or maintenance. The total cost can be expressed as $C_1 + C_2$, where $C_1 = \sum_{j \in V \setminus \{0\}} \sum_{p \in P_1} c_1 x_{0j}^p + \sum_{j \in V \setminus \{0\}} \sum_{p \in P_2} c_2 x_{0j}^p$ and $C_2 = \sum_{i \in V} \sum_{j \in V} \sum_{p \in P_1 \cup P_2} cl_{ij} x_{ij}^p$.

As for the customer satisfaction objective, it is primarily influenced by the total waiting time T for maintenance of all faulty charging spots and the duration for which the working station lacks a professional engineer, denoted as T' . T is calculated as the sum of repair completion times for each faulty charging spot, thus, $T = \sum_{i \in V''} \max_{p \in P_2} t_i^p$. T' represents the earliest return time of the professional engineers to the working station, thus, $T' = \min_{p \in P_2} \max_{i \in V' \cup V''} (t_i^p + \tau_{i0}) x_{i0}^p$. In summary, the following definition can be given.

Definition 1: The loss of composite customer satisfaction is expressed as $\sigma sT + s'T'$, where σ is the impact of specific scenarios related to the fluctuation of charging demand.

Considering both objectives of the problem and all the constraints, the routing optimization problem for large-scale inspection-maintenance of EV charging spots with multiple types of personnel and composite customer satisfaction is formulated as follows.

$$\min f_1 = C_1 + C_2 \quad (1)$$

$$\min f_2 = \sigma sT + s'T' \quad (2)$$

$$\text{s.t.} \quad \sum_{j \in V} x_{ji}^p = \sum_{j \in V} x_{ij}^p \quad \forall i \in V; p \in P_1 \cup P_2 \quad (3)$$

$$\sum_{p \in P_1 \cup P_2} y_i^p = 1 \quad \forall i \in V \setminus \{0\} \quad (4)$$

$$\sum_{j \in V} x_{ij}^p = y_i^p \quad \forall i \in V; p \in P_1 \cup P_2 \quad (5)$$

$$\sum_{p \in P_1 \cup P_2} y_0^p \leq q_1 + q_2 \quad (6)$$

$$t_0^p = 0 \quad \forall p \in P_1 \cup P_2 \quad (7)$$

$$t_i^p + \tau_{ij} + e_j - t_j^p \leq (1 - x_{ij}^p)M \quad \forall i \in V; j \in V \setminus \{0\}; p \in P_1 \cup P_2 \quad (8)$$

$$\sum_{i \in V''} y_i^p / M \leq v^p \quad \forall p \in P_1 \cup P_2 \quad (9)$$

$$\sum_{p \in P_1 \cup P_2} v^p \leq q_2 \quad (10)$$

$$\sum_{i \in S} \sum_{j \in S} x_{ij}^p \leq |S| - 1 \quad (11)$$

$$\forall S \subseteq V \setminus \{0\}; |S| \geq 2; p \in P_1 \cup P_2$$

$$C_1 = \sum_{j \in V \setminus \{0\}} \sum_{p \in P_1} c_1 x_{0j}^p + \sum_{j \in V \setminus \{0\}} \sum_{p \in P_2} c_2 x_{0j}^p \quad (12)$$

$$C_2 = \sum_{i \in V} \sum_{j \in V} \sum_{p \in P_1 \cup P_2} cl_{ij} x_{ij}^p \quad (13)$$

TABLE 2. Notation of parameters.

Parameter	Notation
$G(V, A)$	The network.
$V = \{0\} \cup V' \cup V''$	V is the set of vertices, where $\{0\}$ represents the working station, $V'(V' = n)$ represents the set of charging spots requiring inspection, $V''(V'' = m)$ represents the set of charging spots requiring maintenance.
$A = \{a_{ij} \mid i, j \in V\}$	The set of arcs in the network.
P_1	The set of general personnel.
P_2	The set of professional engineers.
q_1	The maximum quantity of general personnel that can be arranged to the working station.
q_2	The maximum quantity of professional engineers that can be arranged to the working station.
l_{ij}	The length of arc a_{ij} .
τ_{ij}	The passing time of arc a_{ij} .
e_i	The time required to inspect or maintain charging spot i .
c	The cost incurred by each personnel for traveling a unit distance.
c_1	The fixed cost associated with arranging general personnel.
c_2	The fixed cost associated with arranging professional engineers.
s	The loss of customer satisfaction per unit waiting time for maintenance of each faulty charging spot.
s'	The loss of customer satisfaction per unit time when the working station lacks a professional engineer on site.

The objective function (1) represents the minimization of the total cost. The objective function (2) represents the minimization of the loss of composite customer satisfaction. Constraint (3) ensures that the number of personnel entering and leaving each vertex is equal. Constraint (4) guarantees that each charging spot is visited only once by either type of personnel. Constraint (5) indicates that personnel can inspect or maintain a charging spot only if their route passes through that charging spot. Constraint (6) represents the limitation on the total number of available personnel. Constraints (7)-(8) describe the temporal relationship during the inspection-maintenance process, where M is a large positive number. Constraint (9) determines whether the route of a personnel includes a charging spot requiring maintenance. Constraint (10) represents the limitation on the number of available professional engineers. Constraint (11) prevents the formation of sub-tours, where S is a subset of charging spots. Constraints (12)-(13) are the expressions for the costs involved. Constraint (14) represents the total waiting time for maintenance of all faulty charging spots. Constraint (15) represents the duration for which the working station lacks a professional engineer. Constraints (16)-(17) define the domains of the decision variables.

IV. MULTI-STAGE SOLUTION APPROACH

Building upon the routing optimization model for large-scale inspection-maintenance of EV charging spots presented earlier, we now propose a solution approach. While it is possible to solve this model directly, the sheer number of charging spots in a region renders this approach computationally infeasible. To address this challenge, we have developed a multi-stage solution approach, consisting of three stages, as shown in Figure 3. In Stage A, we implement K -means clustering to reduce the scale of the proposed problem. In Stage B, we estimate the cost and time for each cluster using a fast greedy approach. In Stage C, we employ an adaptive NSGA-II to solve the reduced-scale model. This generates a Pareto frontier from which the decision-maker can select a preferred solution.

A. STAGE A: REDUCING THE SCALE OF THE PROPOSED PROBLEM

In Stage A, we address the challenge posed by the large number of charging spots in a region by reducing the scale of the proposed problem. Considering that only a small proportion of charging spots typically require maintenance, and maintaining a charging spot takes significantly longer than inspecting one, we retain all the charging spots requiring maintenance from the original problem and cluster the charging spots requiring inspection, thus forming several clusters. By using the centroid of each cluster as the sole charging spot to replace all the charging spots in this cluster, we reduce the scale of the proposed problem. Here, the widely used K -means clustering is utilized [37]. It is worth noting that the value of K is determined based on the relationship between

$$T = \sum_{i \in V''} \max_{p \in P_2} t_i^p \quad (14)$$

$$T' = \min_{p \in P_2} \max_{i \in V' \cup V''} (t_i^p + \tau_{i0}) x_{i0}^p \quad (15)$$

$$t_i^p \geq 0 \quad \forall i \in V; p \in P_1 \cup P_2 \quad (16)$$

$$x_{ij}^p, y_i^p, v^p \in \{0, 1\} \quad \forall i, j \in V; p \in P_1 \cup P_2 \quad (17)$$

K and m . Our goal is to reduce the total number of charging spots after clustering to N , which ensures reasonable computation time in the subsequent stages where we employ the adaptive NSGA-II algorithm. Therefore, it can be inferred that $K = N - m$. Afterward, K -means clustering can be applied.

B. STAGE B: ESTIMATING OF THE COST AND TIME FOR EACH CLUSTER

After Stage A, we can obtain K clusters of charging spots. Each cluster is represented by its centroid, effectively replacing all charging spots within this cluster. Consequently, we obtain a new network $G^R(V^R, A^R)$, where V^R includes each charging spot requiring maintenance and the centroid of each cluster, and A^R represents the set of arcs in $G^R(V^R, A^R)$. While this simplified network reduces the overall problem scale, applying the proposed model directly to $G^R(V^R, A^R)$ presents two key challenges. Although the time spent inspecting a single charging spot is relatively short, the cumulative time required to inspect multiple spots within each cluster becomes significant and cannot be ignored. Furthermore, there is an associated travel cost for personnel conducting inspections within each cluster.

We propose a fast greedy method here to estimate the cost and time for each cluster. Let V_k^R denote the set of charging spots in cluster k ; e_k represent the total time cost for inspecting cluster k ; c_k indicate the total traveling cost for inspecting cluster k . We propose an estimating approach. First, we need to find the traveling salesman problem (TSP) route within V_k^R , which will form a circle path, as shown in Figure 4. This TSP route can be recorded as TSP_k . Second, the total traveling cost of TSP_k can be recorded as c_k . Next, let the sum of e_i for all charging spots in V_k^R plus the total traveling time of TSP_k be e_k . Finally, the objective (1) can be replaced with $\min f_1 = C_1 + C_2 + \sum_{k \in K} c_k$.

Special attention should be paid to how to compute TSP_k . Considering the potential for a large number of clusters, which necessitates numerous algorithm runs, and the need for a fast approach to handle the scale of the problem, we choose Christofides Algorithm to compute the TSP routes efficiently. The Christofides Algorithm offers a good balance between solution quality and computational efficiency, with a time complexity of $O(n^3)$. Thus, we have the following proposition.

Proposition 1: The time complexity of the entire Stage B is $O(Kn^3)$.

Generally speaking, the value of K is chosen to be of a reasonable order of magnitude to ensure the efficiency of the solution. Therefore, the process of Stage B can be completed in polynomial time.

C. STAGE C: SOLVING THE REDUCED-SCALE MODEL

Having completed Stage A and Stage B, we now face the challenge of obtaining the Pareto frontier for the bi-objective

optimization model. This task is inherently more complex than solving a single-objective model [38], [39], [40]. To address this complexity, we employ NSGA-II, which is widely recognized for its strong search capabilities in solving bi-objective routing models [41]. However, NSGA-II is not without limitations, particularly its tendency to coverage on local optima. To mitigate this issue, we implement an adaptive NSGA-II. This adaptive approach enhances the algorithm's ability to explore the solution space more thoroughly, increasing the likelihood of finding global optima. The procedure of the adaptive NSGA-II is illustrated in Figure 5. Then, explanations of the comparatively important steps are provided in following part.

1) CODING AND DECODING

Positive integers are used for coding. A set of positive integers $\{1, 2, \dots, 2(K + m) - 1\}$ is generated for coding, where K is the number of clusters, and m is the number of charging spots requiring maintenance. The elements in this set are randomly arranged to form an array. For example, as illustrated in Figure 6, consider a case where $K = 3$, $m = 2$. The coding array contains $2(K + m) - 1 = 9$ elements. In this example, the bolded numbers represent the charging spots that require maintenance. During the decoding process, all elements greater than $K + m$ are first changed to 0. Then, the array is divided into routes, using 0 as the separators. Finally, the routes are assigned to either general personnel or professional engineers based on whether the route contains charging spots requiring maintenance.

2) ADAPTATIVE CROSSOVER PROBABILITY AND MUTATION PROBABILITY

The standard NSGA-II usually adopts fixed crossover and mutation probability. However, maintaining constant probability throughout the entire solving process can lead to many issues. For example, lowering probability may hinder the generation of new and superior individuals, while higher probability can make convergence difficult. From the perspective of the overall population evolution process, the crossover and mutation probability should gradually decrease and eventual stabilization. This adaptive approach allows for more exploration in the early stages and better convergence in later stages. To address this, we propose an adaptive model for crossover and mutation probability. p_c^0 and $p_c^{\min}$ represent the initial crossover and mutation probability respectively; p_m^0 and $p_m^{\min}$ represent the lower bounds of the crossover and mutation probability respectively; δ is the number of iterations; a, b, c, d are parameters required to implement the adaptive probability.

$$p_c = \begin{cases} p_c^0 a^{-\delta/b} & p_c \geq p_c^{\min} \\ p_c^{\min} & p_c < p_c^{\min} \end{cases} \quad (18)$$

$$p_m = \begin{cases} p_m^0 c^{-\delta/d} & p_m \geq p_m^{\min} \\ p_m^{\min} & p_m < p_m^{\min} \end{cases} \quad (19)$$

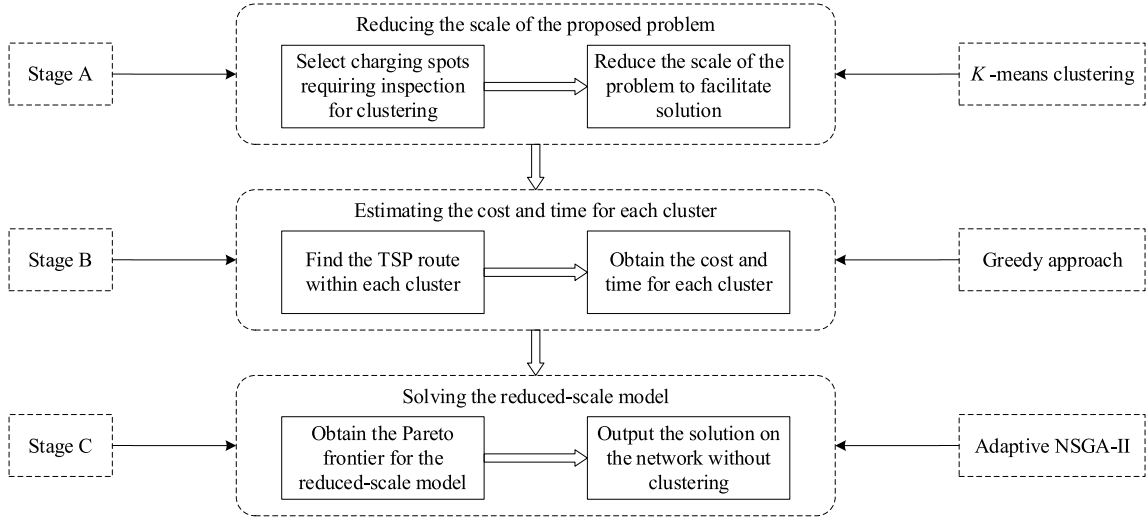


FIGURE 3. Procedure of the multi-stage approach.

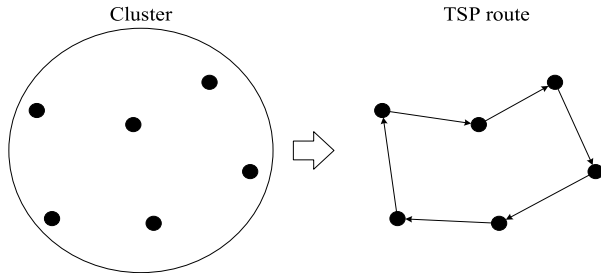


FIGURE 4. TSP route in a cluster.

3) OUTPUT THE SOLUTION ON THE NETWORK WITHOUT CLUSTERING

By applying the adaptive NSGA-II above, we can obtain the Pareto frontier, from which the EV charging spot operator can choose a preferred solution. However, it should be noted that this chose solution is based on the clustered network $G^R(V^R, A^R)$, and we still need to output the solution on the network without clustering. We propose a method of breaking TSP and reconnecting them. First, choose any arc on TSP_k , break the arc and connect the two endpoints of this arc to the arc of the in/out cluster k , as shown in Figure 7. Then, record the cost of breaking this arc and choose another arc to repeat the above process. The arc corresponding to the lowest cost scenario is ultimately chosen to perform the breaking TSP and connecting.

The time complexity of breaking each arc is $O(1)$. Since there are at most n arcs in a TSP route, the time complexity of handling each cluster is $O(n)$. Thus, we have the following proposition, indicating that this method operates in polynomial time.

Proposition 2: The time complexity of outputting the solution on the network without clustering is $O(Kn)$.

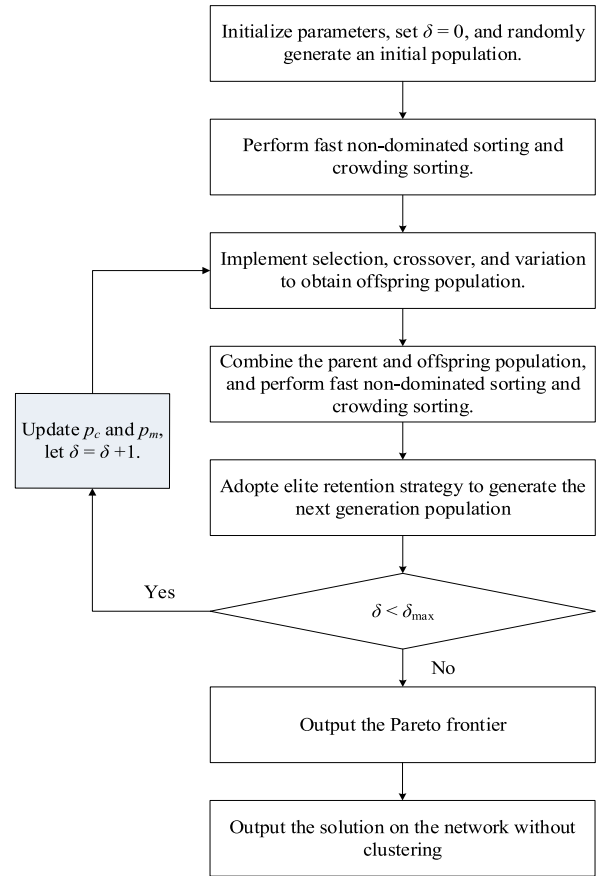


FIGURE 5. Procedure of the adaptive NSGA-II.

V. NUMERICAL RESULTS

This section presents the experimental validation and analysis of our proposed model and solution approach. First, we begin

B. COMPARISON WITH THE MODEL IN PREVIOUS STUDIES

To verify the effectiveness of the proposed model in this paper, we compare our solution results with those obtained from models in previous studies. Our paper develops a bi-objective model that considers multiple types of personnel and composite customer satisfaction, whereas previous research typically focused on single-objective models aimed at cost minimizations and considered only one type of personnel. The solution results are presented in Table 5. It can be observed that the proposed model in this paper yields superior solution. In terms of customer satisfaction, our results consistently outperform those obtained from models in previous studies, showing an average improvement of 49.16%. This demonstrates that the proposed model in this paper can more effectively solve the routing optimization problem for inspection-maintenance of EV charging spots, significantly reducing the loss of composite customer satisfaction.

C. SENSITIVITY ANALYSIS

1) ANALYSIS OF DIFFERENT PROFICIENCY OF PROFESSIONAL ENGINEERS

Professional engineers play a vital role in the proposed problem, as their attributes significantly influence the maintenance of charging spots and customer satisfaction, thereby significantly impacting the routing plan. In practical scenarios, charging spot operators employ professional engineers with diverse characteristics. Experienced professional engineers can accomplish the maintenance of charging spots in a shorter time, albeit typically commanding higher salaries. Conversely, novice professional engineers may require more time to complete the maintenance task but generally receive relatively lower salaries.

To compare the impact of professional engineers with diverse attributes, we prepare three distinct parameter combinations. The first combination is the original one, where $e_i = e_i (i \in V'')$, $c_2 = c_2$. The second combination corresponds to experienced professional engineers, with $e_i = 0.8e_i (i \in V'')$, $c_2 = 1.1c_2$. The third combination represents novice professional engineers, with $e_i = 1.2e_i (i \in V'')$, $c_2 = 0.9c_2$. By running the program with these parameter combinations, we obtain the Pareto frontiers displayed in Figure 10.

Figure 10 reveals an intriguing phenomenon that contradicts initial intuition. The second combination, representing experienced professional engineers, leads to a significant cost reduction despite their relatively higher salaries. These counterintuitive results can be attributed to the increased efficiency of experienced professional engineers in maintaining charging spots, which improves the overall inspection-maintenance process. Importantly, this finding indicates that experienced professional engineers does not increase the total cost, as one might expect due to their higher salaries. Instead, their expertise actually reduces the total cost. This underscores the crucial role of experienced professional engineers in inspection and maintenance operations. Based on these results,

it is evident that charging spots operator should maintain a certain number of experienced professional engineers in their workforce.

2) ANALYSIS OF DIFFERENT NUMBER OF PERSONNEL

The routing for inspection-maintenance of EV charging spots is constrained by the number of personnel. In this study, we analyze the impact of varying number of personnel on the results. We consider two types of personnel: general personnel and professional engineers. Our analysis focuses on two different scenarios: reducing only the number of professional engineers, simultaneously reducing the number of general personnel and professional engineers. The results for these different scenarios are presented in Figure 11.

As shown in Figure 11, reducing the number of professional engineers leads to a deterioration of the Pareto frontier and a significant decrease in the number of generated solutions. This indicates that the number of professional engineers is crucial. With only a small number of professional engineers, achieving efficient routing for inspection-maintenance of EV charging spots becomes more challenging. When both the number of general personnel and professional engineers are reduced, the Pareto frontier further deteriorates. This demonstrates that we cannot underestimate the importance of general personnel, as they carry out most of the inspection work. Maintaining an adequate number of general personnel to meet inspection needs is equally crucial for the overall efficiency of the system.

3) ANALYSIS OF DIFFERENT TRAVELLING SPEEDS

The inspection-maintenance of EV charging spots is typically conducted in urban areas, necessitating a discussion on the impact of traffic conditions on the results. Traffic conditions primarily affect the travelling speeds of personnel, which in turn influence the time spent traversing the network, ultimately impacting the value of the objective function. To analyze this impact, we generated two different scenarios. In the first scenario, we increased the travelling speeds of all personnel by 20% to simulate improved traffic conditions. In the second scenario, we increased both the travelling speeds of all personnel and the unit cost by 20% to simulate the adoption of more efficient transportation methods. The results of these simulations are presented in Figure 12.

Figure 12 demonstrates that in most cases, increasing travelling speeds can improve the Pareto frontier, resulting in a reduction of both the total cost and the loss of composite customer satisfaction. However, when the increase in travelling speeds is accompanied by an increase in unit costs, the conclusion is the opposite. This indicates that relying solely on more efficient transportation methods does not necessarily lead to an improved solution. In general, EV charging spot operators should prioritize conducting inspection-maintenance during periods of favorable traffic conditions, rather than employing more expensive and efficient transportation methods during times of poor traffic flow.

TABLE 4. Some indicators for the pareto frontiers.

	Multi-stage approach					Global optimization		
	Number of solutions	Average total cost	Gap	Average loss of composite customer satisfaction	Gap	Number of solutions	Average total cost	Average loss of composite customer satisfaction
1	7	674.61	32.05%	876.97	-11.40%	3	510.89	989.84
2	6	592.05	27.37%	1125.49	-43.91%	5	464.84	2006.53
3	7	859.59	32.18%	1138.27	2.42%	20	650.33	1111.38
4	11	756.41	23.31%	1233.80	-19.97%	12	613.42	1541.72
5	18	1196.73	-13.59%	1280.65	16.95%	26	1384.94	1095.02
6	23	1297.73	-13.04%	2395.97	-0.50%	43	1492.34	4791.91
7	49	2041.35	-28.10%	1621.83	19.65%	19	2839.11	1355.49
8	57	3281.62	-9.97%	2023.58	40.70%	28	3644.92	1438.24
Average gap			6.28%		0.49%			

TABLE 5. Comparisons with the model in previous studies.

Instance	Average loss of composite customer satisfaction	Loss of composite customer satisfaction of the model in previous studies	Improvement of customer satisfaction
1	876.97	2082.39	57.89%
2	1125.49	3360.74	66.51%
3	1138.27	1993.60	42.90%
4	1233.80	2518.44	51.01%
5	1280.65	2830.27	54.75%
6	2395.97	7705.66	68.91%
7	1621.83	2277.26	28.78%
8	2023.58	2611.87	22.52%
Average			49.16%

4) ANALYSIS OF DIFFERENT WEIGHT OF CUSTOMER SATISFACTION

One of the model's objective functions is to minimize the loss of composite customer satisfaction, which is composed of two types of customer satisfaction. The parameter σ is the weight used to combine these satisfaction measures. The weight σ reflects the impact of specific scenarios on the overall customer satisfaction. In this section, we conduct a sensitivity analysis on σ . We set $\sigma = 1$, $\sigma = 0.5$, and $\sigma = 1.5$, and solve the model for each case. The results of this analysis are presented in Figure 13.

Figure 13 illustrates the impact of varying σ on the Pareto frontier. As σ increases, we observe a gradual rise in the loss of composite customer satisfaction. Conversely, when σ decreases, the loss of composite customer satisfaction generally declines in most instances. However, Instance 7 presents an anomaly where the loss increases despite a lower σ . These

observations lead to several insights. First, the weight of customer satisfaction has a significant impact on the overall results. Second, when the loss of customer satisfaction caused by waiting for maintenance of faulty charging spots is relatively important, there is a need to focus more on customer satisfaction. In cases where the loss of customer satisfaction caused by "absence of professional engineer" is more critical, the approach needs to be tailored to specific circumstances. Overall, the analysis suggests that waiting time for maintenance of faulty charging spots has a higher importance for customer satisfaction.

D. CASE STUDY

In this section, we apply the aforementioned model and solution approach to address the inspection-maintenance of charging spots in Xi'an, one of the major cities in China. Xi'an's significance in the EV market is evident from its production of 983,800 EVs in 2023, with the city's total number of motor vehicles reaching 5,297,300 by the end of the year. Through collaboration with the Xi'an Transportation Development Research Center and Xi'an Transportation Information Center, we obtained the layout of existing charging spots in Xi'an, which includes the coordinate information for 1,281 charging spots. Table 6 presents the coordinate information for a sample of 10 out of the total 1,281 charging spots. By plotting all 1,281 charging spots. Figure 14 provides a visual representation of the charging spot data. In Figure 14(a), we plot the charging spots to show their distribution across the city. Figure 14(b) offers a more detailed view, where the horizontal axis represents longitude and the vertical axis represents latitude, the red points indicate charging spots requiring inspection, while the blue points represent charging spots requiring maintenance.

We obtained empirical values for the model parameters through interviews with TELD (a major electric vehicle

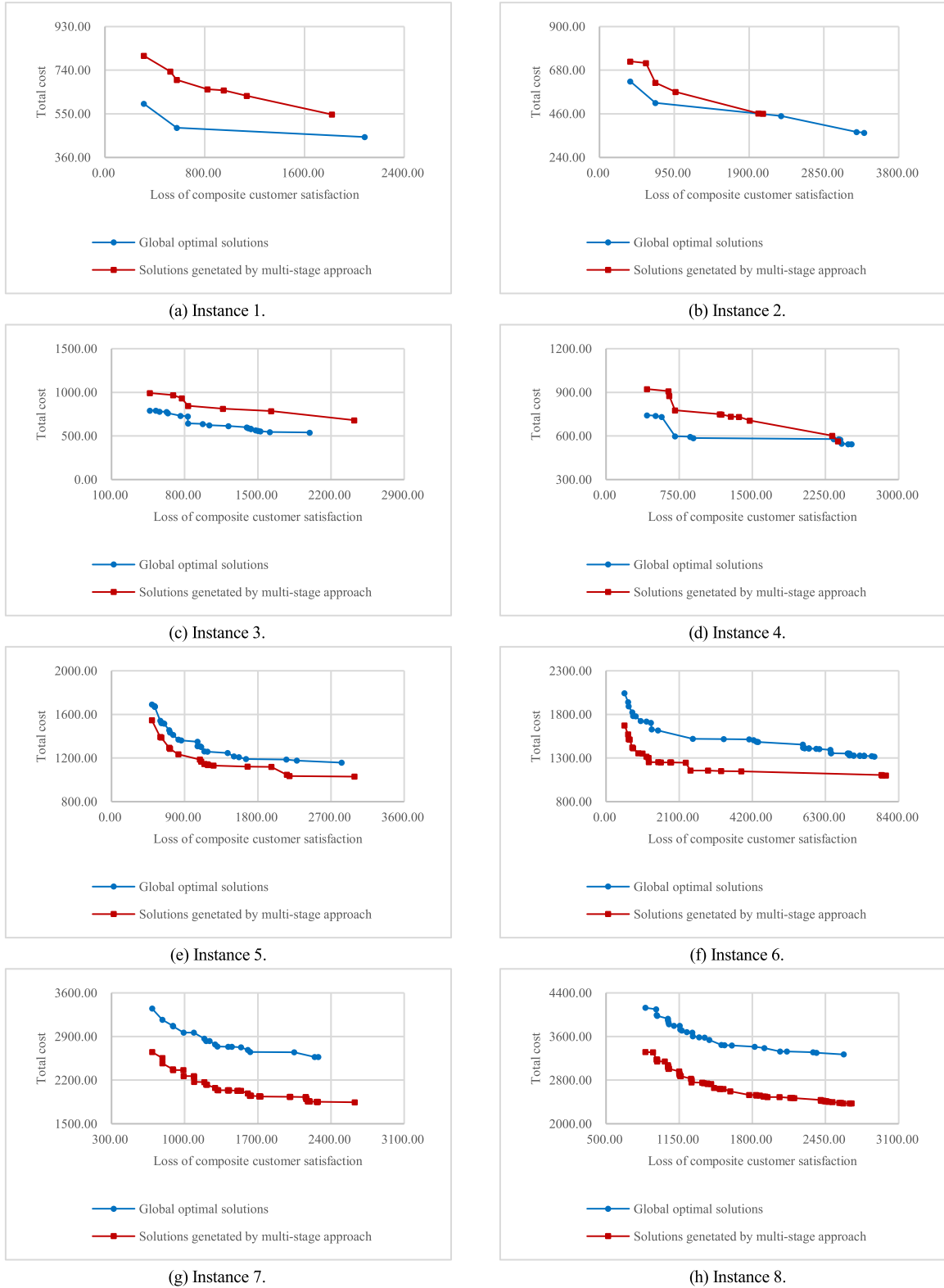


FIGURE 9. Generated Pareto frontiers of each instance.

charging network operator in China). Specifically, we set $c = 3\text{CNY}$, $c_1 = 100\text{CNY}$, $c_2 = 500\text{CNY}$, and $e_i = 1 \text{ min} (i \in V')$. For $e_i (i \in V'')$, its values are taken randomly in the range of $[50 \text{ min}, 70 \text{ min}]$. For $\tau_{ij} (i, j \in V)$, its values

are taken by dividing l_{ij} to randomly generated speeds in the range of $[50\text{km/h}, 70\text{km/h}]$. Without loss of generality, we set $q_1 = q_2 = 10$, $s = 1$, $\sigma = 1$, and randomly select $m = 50$ of the charging spots as those requiring maintenance.

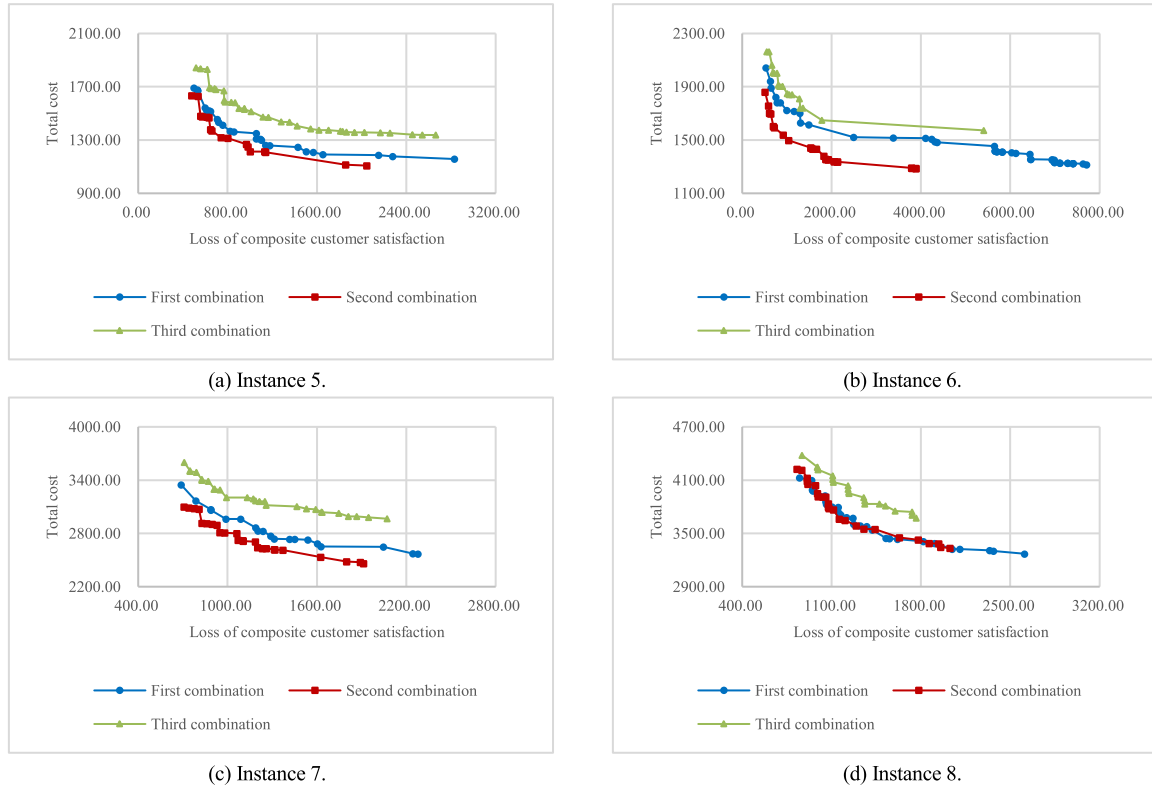


FIGURE 10. Generated Pareto frontiers with different proficiency of professional engineers.

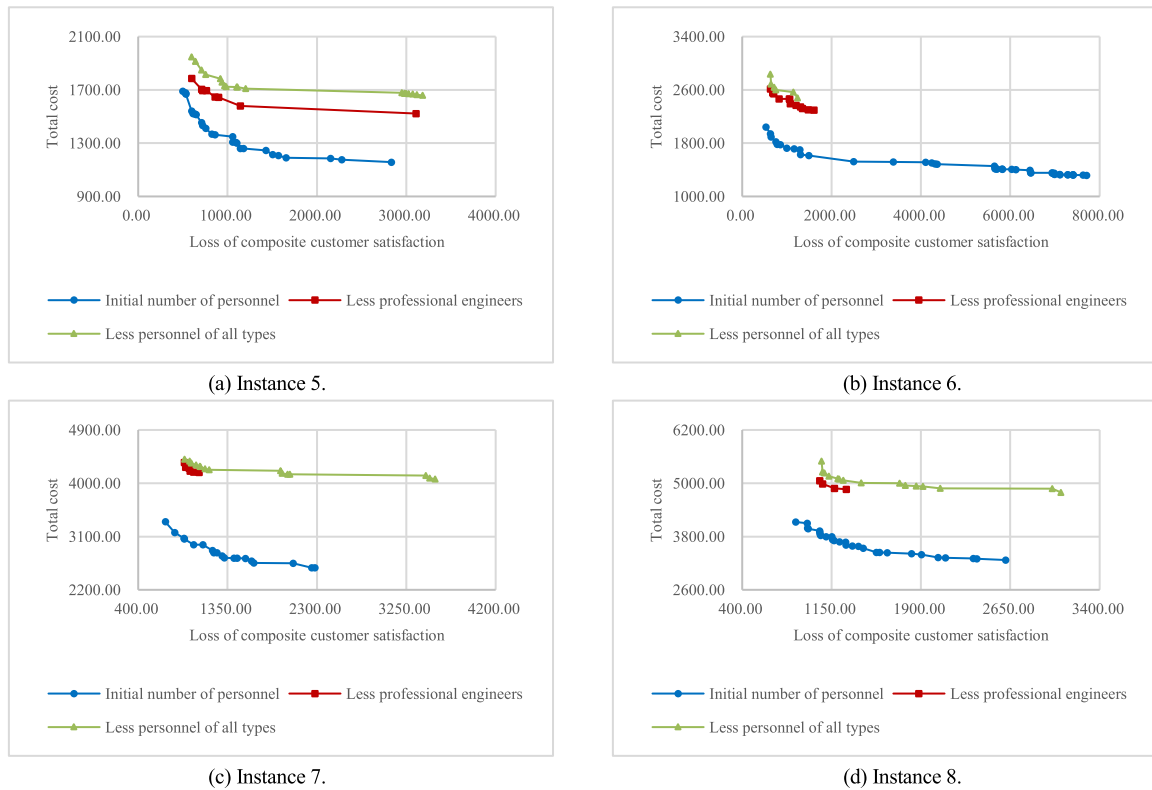
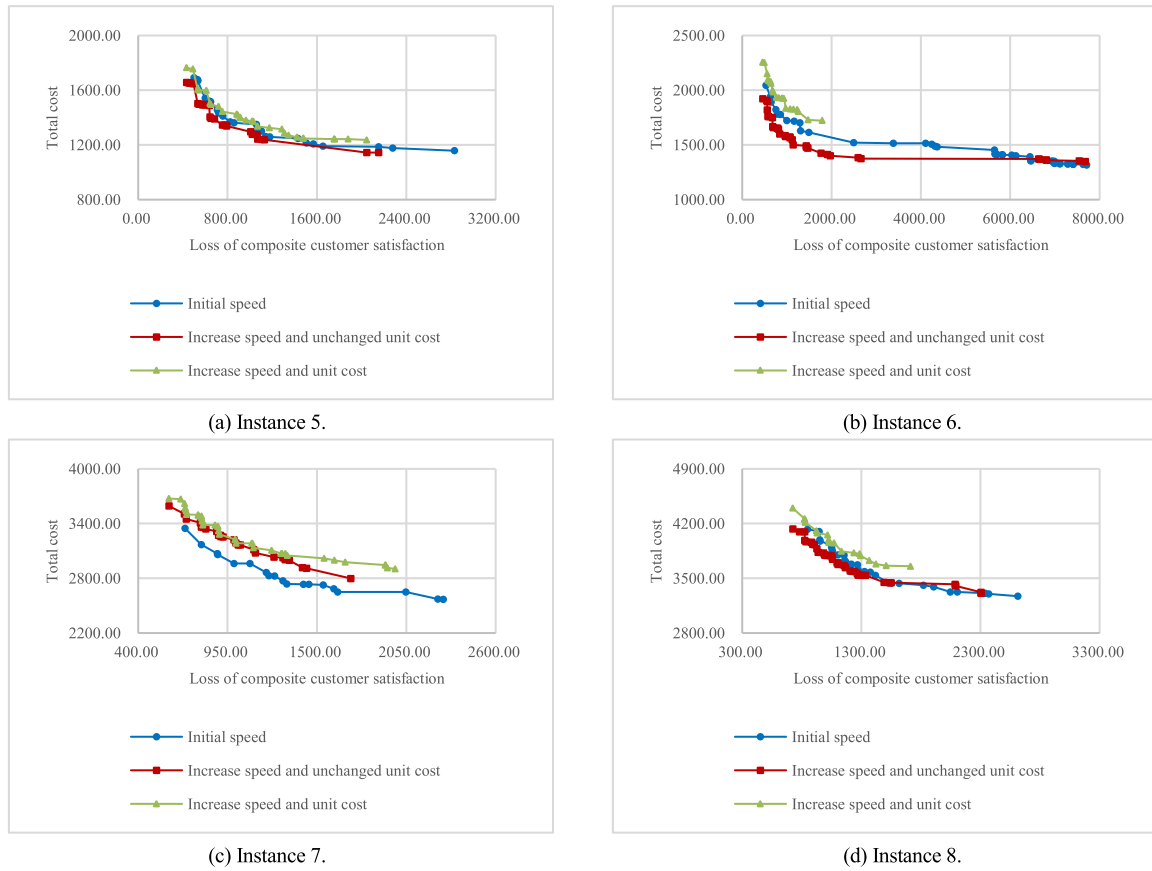
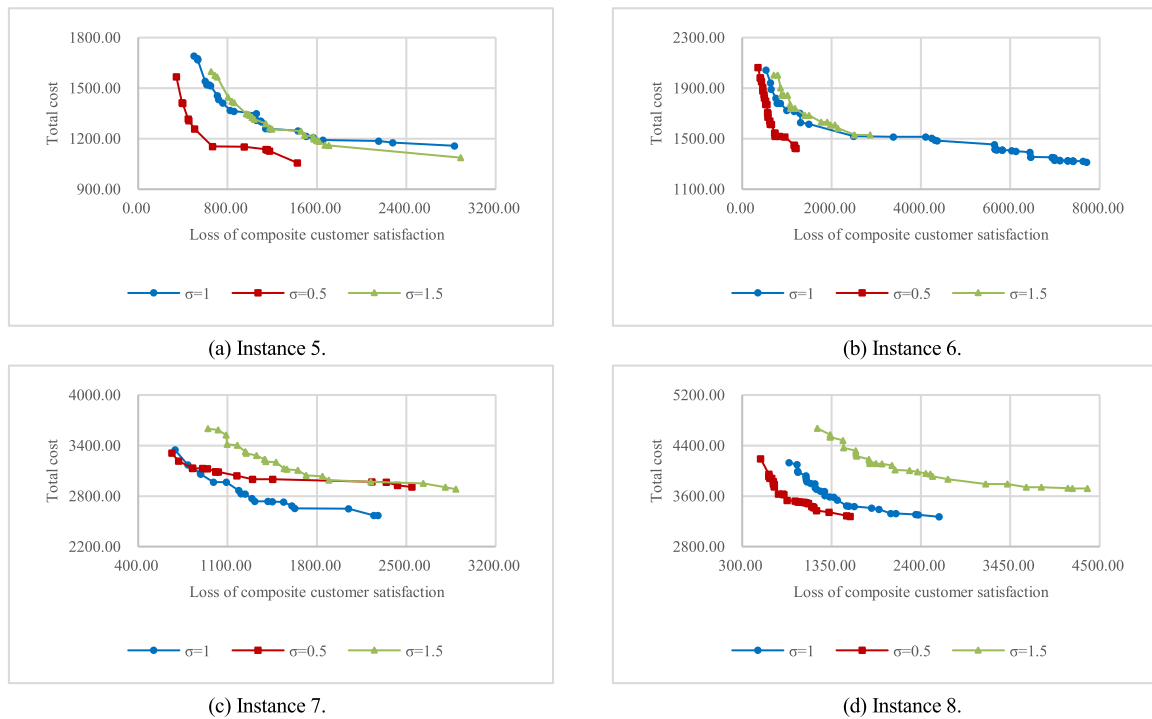


FIGURE 11. Generated Pareto frontiers with different number of personnel.

Due to the large scale of the problem involving 1,281 charging spots, we implemented a reduction strategy to make

the computational process more manageable. We choose an appropriate scale of $N = 100$, resulting in $K =$

**FIGURE 12.** Generated Pareto frontiers with different travelling speeds.**FIGURE 13.** Generated Pareto frontiers with different weight of customer satisfaction.

$N - m = 50$. To achieve this reduction, we perform K -means clustering on the 1,231 charging spots requiring

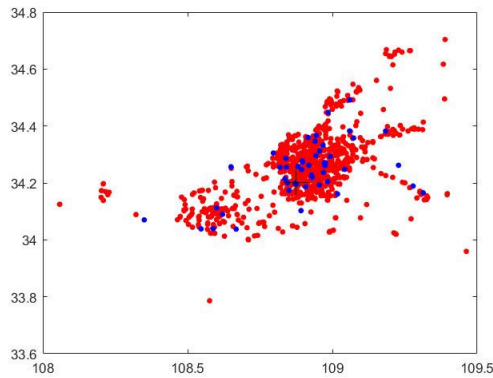
inspection, and the results are shown in Figure 15. Subsequently, we estimate the cost and time for each cluster using

TABLE 6. Coordinate information of partial charging spots.

Name	Longitude (East Longitude)	Latitude (North Latitude)
Zhentaoy charging spots	109.2421	34.2023
StateGrid charging spots	109.2283	34.26198
Jishiyu charging spots	109.3116	34.3872
TELD charging spots	109.3139	34.41365
TELD charging spots	109.2007	34.53219
StateGrid charging spots	108.3503	34.07019
StateGrid charging spots	108.3504	34.07019
YKC charging spots	108.3218	34.08871



(a) Actual layout.

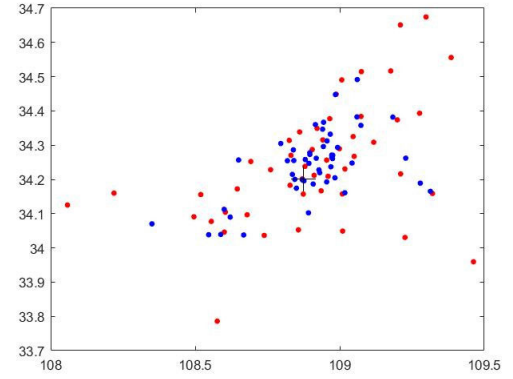
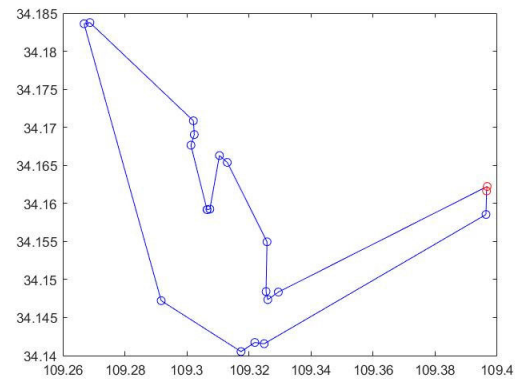
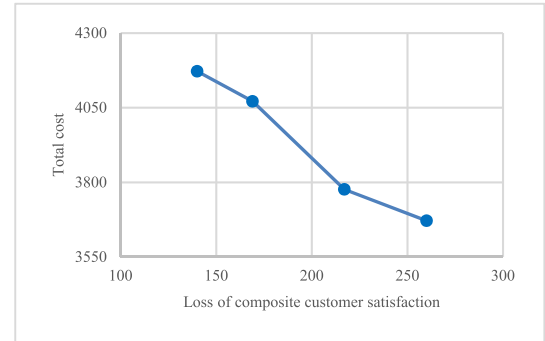


(b) Layout on the coordinate system.

FIGURE 14. Layout of all the 1281 charging spots in Xi'an.

the TSP tour. As a demonstration, Figure 16 displays the TSP route of one of the clusters. Finally, we number the $N = 100$ vertices, with the first 50 requiring inspection and the last 50 requiring maintenance, thus obtaining the network $G^R(V^R, A^R)$.

Subsequently, we employ the adaptive NSGA-II to obtain the Pareto frontier. First of all, we apply the proposed model

**FIGURE 15.** Layout of charging spots after clustering.**FIGURE 16.** TSP route of one of the clusters.**FIGURE 17.** Pareto frontier generated by adaptive NSGA-II.

to $G^R(V^R, A^R)$ by replacing objective (1) with $\min f_1 = C_1 + C_2 + \sum_{k \in K} c_k$ and replacing each e_i in V_k^R by $e_i + e_k$. By solving the problem, the generated Pareto frontier containing the solutions is shown in Figure 17. As a demonstration, we select a representative solution from the Pareto frontier and present its corresponding in Figure 18. The results indicate that our proposed model and multi-stage approach can effectively solve practical routing optimization problems for the inspection-maintenance of EV charging spots with multiple types of personnel and composite customer satisfaction.

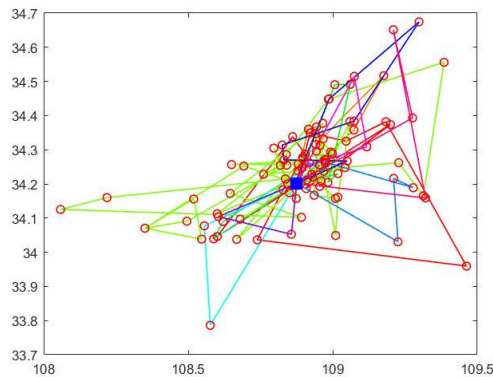


FIGURE 18. Display of the routes of a solution on the Pareto frontier.

VI. CONCLUSION

To alleviate the issue of “range anxiety” experienced by drivers, many countries have constructed extensive EV charging spot systems with a large number and wide coverage areas. These charging spots require regular inspections to detect potential safety hazards, and the charging piles exhibit a significant failure rate, necessitating frequent maintenance. Besides, the inspection-maintenance of charging spots possesses some distinct characteristics. The number of charging spots is enormous, and the working station set by the EV charging spot operators need to undertake customer service in addition to conducting inspection and maintenance. Considering these factors, along with the role of general personnel and professional engineers, we propose a bi-objective routing optimization model. This model aims to simultaneously minimize the cost of arranging personnel and traveling between charging spots, and the loss of composite customer satisfaction. Due to the large scale of the problem, we develop a multi-stage approach. This approach includes reducing the scale of the proposed problem and utilizing an adaptive NSGA-II to generate the Pareto frontier.

The model and proposed multi-stage approach are tested by a series of instances. The results show that the proposed multi-stage approach can effectively solve the problem. Besides, the proposed model in this paper can significantly reduce the loss of composite customer satisfaction comparing with the models in previous studies. Furthermore, the study conducts sensitivity analysis on several parameters. The results show that experienced professional engineers can improve inspection-maintenance efficiency, even with higher wages. The number of personnel is crucial, and it is crucial to ensure a sufficient workforce. It is beneficial to reduce travel time between charging spots, unless it leads to increased traveling costs. The weight assigned to customer satisfaction has a notable impact, but its optimal value needs to be determined based on specific circumstances. Finally, the study presents a case study addressing the large-scale inspection-maintenance of EV charging spots in Xi'an, China.

Future research can be conducted in the following aspects. First, the present study assumes that all personnel depart from the working station at the beginning. Future research

can investigate scenarios where the departure time is a decision variable. Second, it is necessary to conduct research on routing optimization for inspection-maintenance of EV charging spots under conditions with time windows. Third, given that these operations typically occur in urban environment with significant traffic fluctuations, future studies could explore optimization based on time-dependent network. Fourth, given the large scale of the problem, more effective solution approaches can be designed to address this problem.

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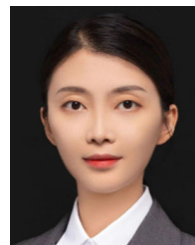
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